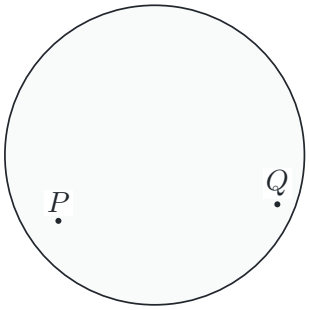
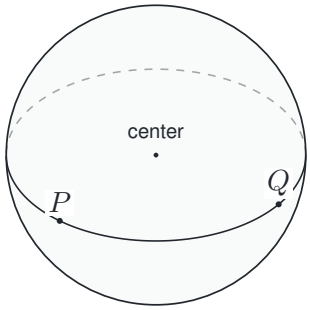
Keep the ball on its support. Keep the string against its surface and its endpoints on the same two markers. Dashed arcs lie on the back of the ball.

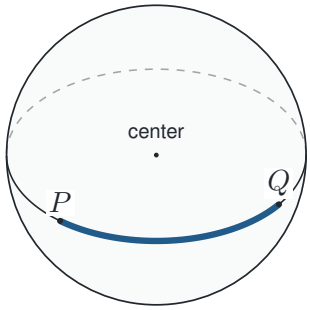
A *great circle* is the circle made by a flat cut through the ball's center. A shortest surface route between two points runs along a great circle through them.



Mark P and Q



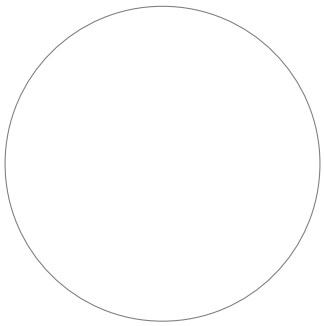
Their great circle



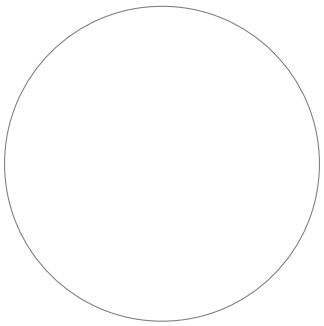
Shorter arc: P to Q

Problem 1: Choose a close pair of markers, a far-apart pair that is not opposite, and a pair exactly opposite each other on your ball. Find the shortest surface routes for each pair. Which pairs let you find two different routes of the same shortest length? Draw your routes.

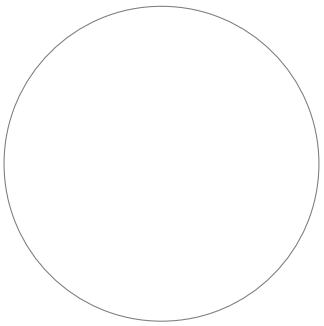
Close pair



Far-apart pair

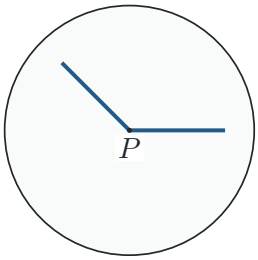


Opposite pair

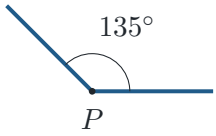


Join three different markers by their shortest surface routes. Do not use opposite markers or put all three on one great circle. Use the smaller region inside the three sides.

An angle on the ball is the angle between the directions of the two sides at their meeting point. Use the inside corner.



Two sides at P

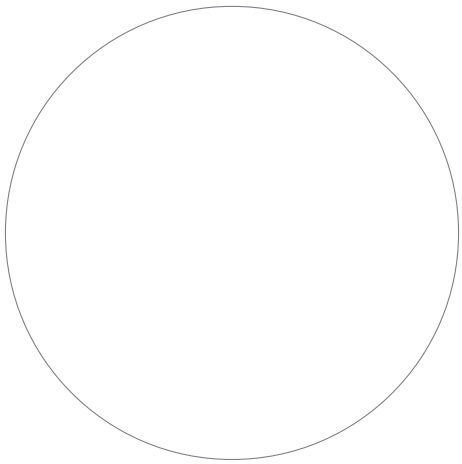


Directions close to P

$P:\ 135°$

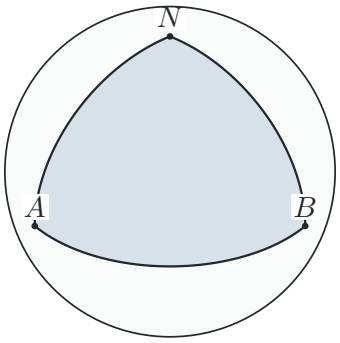
Angle record

Problem 2: Make a triangle on your ball with as many right-angle corners as you can. Check its corners with the paper right angle. Can all three corners be right angles? Draw your best triangle and mark its corners.



The equator is a great circle. N is the pole above it. A meridian is a great-circle semicircle from N to the opposite pole.

Problem 3: Keep N fixed. Move A and B on the equator, using its shorter arc as the third side. Make triangles whose corner at N is smaller than a right angle, equal to a right angle, and larger than a right angle. Draw each triangle and mark all its corners. Which changes make the triangle cover more of the ball?



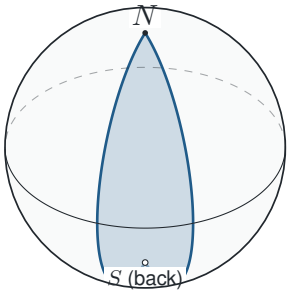
N corner smaller than a right angle

N corner equal to a right angle

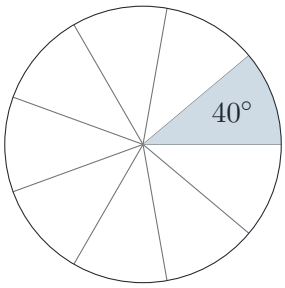
N corner larger than a right angle

Extra workspace

A *lune* is the strip between two great-circle semicircles from N to its opposite pole S. The strip's angle is the gap between them at N. Fractions refer to the ball's surface.



$40°$ lune



View down toward N

9 equal lunes fill the ball

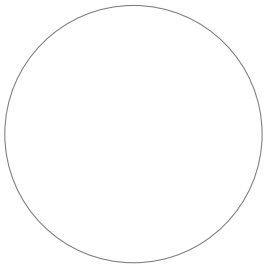
$360 \div 40 = 9$

One lune: $\frac{1}{9}$ of the surface

Area record

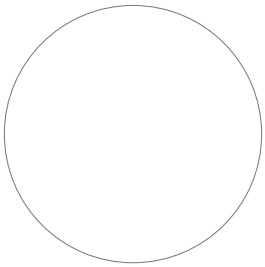
Problem 4: Use the marked meridian pairs with gaps $45°$, $60°$, and $120°$. For each pair, make a triangle with N and the two equator points. What fraction of the ball's surface does each triangle cover?

$45°$ gap



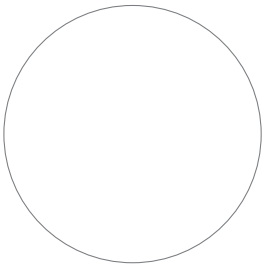
Fraction: _____

$60°$ gap



Fraction: _____

$120°$ gap



Fraction: _____